Computer Vision for Developers | أكاديمية سدايا ▪ SDAIA ACADEMY

Formula 1 Racing Cars Tracking Using Deep Learning

Developing an intelligent system for fast object detection and tracking using advanced YOLOv8

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Project Scope / 01

Project Idea & Application Domain

Deep Learning

Leveraging pre-trained models to solve real-world computer vision challenges.

Video Analysis

Processing video frames and extracting clear, reliable visual outputs.

Vehicle Tracking

Precise monitoring of racing cars trajectory and movement in real-time.

Problem Definition & Expected Output

The Problem

.Manual tracking of fast-moving objects across video frames is inefficient and impractical without automated tools

System Purpose

.Automate detection and tracking processes with high accuracy and speed using modern computer vision

Expected Output

.A processed output video featuring visual bounding boxes and tracking IDs assigned to each vehicle

Implementation & Model Architecture

Environment

Executed and tested all scripts and pipelines within Google Colab cloud .notebooks

YOLOv8 Model

Utilized YOLOv8 for its high inference .speed and exceptional tracking efficiency

Vision Task

Applied standard Object Detection & .Tracking tasks covered in the curriculum

Performance & Future Improvements

Future Improvements

Perform fine-tuning on a custom dataset tailored for racing vehicles to enhance detection accuracy.

Failure Cases

Temporary loss of tracking (occlusion) can occur when vehicles overlap or under extreme camera angles.

Successful Prediction

The model successfully tracked racing cars and assigned tracking IDs throughout the video frames.

Thank You

We welcome your questions and inquiries about the project